

UES-Br and LIBRAS-Mediated Music Learning: Exploratory Qualitative Study of Deaf Adolescents' Interaction with LIBRAS Academy

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Abstract. This paper reports a qualitative exploratory study of how six deaf adolescents, aged 13 to 15, interacted with LIBRAS Academy, a digital interface for music education mediated by Brazilian Sign Language (LIBRAS). The study was conducted on April 6, 2026, at a specialized institutional site for deaf education and hearing, voice, and language support. Data included a structured first-contact testing protocol, observational annotations from individual sessions, open-ended comments, and responses to the Brazilian Portuguese adaptation of the User Engagement Scale (UES-Br). The analysis characterizes interaction patterns, interface barriers, mediation demands, and the limits of interpreting UES-Br responses in this context. Participants completed the activities and displayed different forms of engagement, including reflective inquiry, efficient task completion, self-correction, selective persistence, and image-based progression under reading constraints. Barriers emerged around task onboarding, visual layout, game pace, scroll mechanics, recognition reliability, and abstract questionnaire vocabulary. Correcting negatively worded UES-Br items was essential for interpretation, but the questionnaire remained dependent on mediation and vulnerable to semantic difficulty and possible acquiescence. The paper contributes empirical evidence about LIBRAS-mediated music learning and analyzes how UES-Br behaved under mediated administration with deaf adolescents.

Keywords: Deaf adolescents · LIBRAS · music education · UES-Br · inclusive learning technologies · accessibility evaluation

1 Introduction

For many deaf students, music education is still organized around sounds that are expected to be heard, named, reproduced, and evaluated through hearing-centered pedagogical routines. Yet musical experience can also be approached through visual patterns, rhythm, gesture, color, memory, and shared symbolic activity. Digital learning environments can turn these forms of access into usable

learning pathways, but only when they are designed and evaluated around deaf learners' language practices and access needs. The problem is not simply the absence of music software for deaf students. It is the lack of evidence about how deaf students interpret instructions, negotiate support, and remain engaged with such interfaces when they encounter them in educational settings.

This problem has methodological consequences. If accessible educational interfaces are evaluated mainly through design intentions, technical demonstrations, or generic usability assumptions, researchers may overestimate their accessibility. The presence of sign language, visual elements, or gesture recognition does not guarantee that instructions are clear, that task rhythm fits the user, that visual layout supports interpretation, or that a post-use questionnaire is linguistically accessible. Engagement is also difficult to interpret. A participant may complete tasks efficiently without wanting to continue, persist in one activity while rejecting another, or report positive impressions while requiring extensive mediation. For inclusive learning technologies, these differences matter because they affect not only how users respond to the platform, but also the reliability of the evidence used to improve it.

Relevant prior work addresses parts of this challenge. Benites' research on music education for deaf children proposed and developed LIBRAS Academy, an interface that combines musical notes, colors, Brazilian Sign Language (LIBRAS), and artificial-intelligence-supported gesture interaction [2,3]. As an interactive learning object organized around sign-based multimodal access to musical content, LIBRAS Academy places deaf learners' communicational repertoires at the center of music education rather than treating accessibility as an add-on. Research on assistive educational technology grounded in semiotic theory has examined how images, written language, and LIBRAS signs can be combined in learning software for deaf and hearing children [9,10]. In Human-Computer Interaction (HCI), the User Engagement Scale (UES) and its Brazilian Portuguese adaptation, UES-Br, provide a structured way to describe perceived engagement across dimensions such as focused attention, perceived usability, aesthetic appeal, and reward [8,7]. These contributions leave open an empirical question: how do deaf adolescent users interact with a LIBRAS-mediated music learning interface for the first time, and how should engagement instrument responses be interpreted when the questionnaire requires translation and adult support?

The research gap addressed in this paper is twofold. Empirically, there is still limited evidence, especially in the Brazilian and Latin American context, on how deaf students interact with sign-language-mediated music learning interfaces in first-contact sessions. Methodologically, little is known about how UES-Br responses should be interpreted when respondents are deaf adolescents and the questionnaire requires translation, explanation, or adult mediation. This paper does not claim to measure learning gains, validate UES-Br for this population, or provide statistical evidence of platform effectiveness. Instead, it offers an exploratory, qualitative account of interaction and engagement in first-contact sessions.

The study was conducted on April 6, 2026, at the Children’s Hearing Center (CeAC), within the Division of Education and Rehabilitation of Communication Disorders (DERDIC), at the Pontifical Catholic University of São Paulo (PUC-SP), in São Paulo, Brazil. Six deaf adolescents aged 13 to 15 participated in individual first-contact sessions with LIBRAS Academy, followed by a mediated UES-Br questionnaire. The analysis is guided by three research questions:

- RQ1: How do deaf adolescents aged 13 to 15 interact with LIBRAS Academy in individual first-contact sessions?
- RQ2: What forms of engagement are suggested by triangulating behavioral observation and UES-Br responses?
- RQ3: How do mediation conditions and linguistic accessibility affect the interpretability of UES-Br responses in this context?

The remainder of the paper is organized as follows. Section II reviews related work on DHH technologies, LIBRAS-mediated music learning, engagement measurement, and accessible evaluation methods, building toward the gap the study addresses. Section III describes the study context, participants, protocol, UES-Br scoring, and qualitative analysis procedure. Section IV reports the results, beginning with observed interaction patterns and interface barriers before treating UES-Br as complementary evidence. Section V discusses the empirical and methodological implications of the findings. Section VI addresses limitations and directions for future work. Section VII concludes the paper.

2 Related Work

Music education for deaf learners has long been designed around hearing-centered assumptions, yet deaf individuals can perceive rhythm, vibration, and visual patterns in music, and digital interfaces can extend those channels of access. Benites addressed this design challenge through a research program that produced LIBRAS Academy, a music learning environment in which notes are associated with colors, handshapes, and gesture-recognition feedback mediated by LIBRAS [2,3]. In this approach, musical notes are not only heard; they are seen, enacted, named, and compared through sign-based interaction, allowing a learner without residual hearing to navigate the interface through visual and LIBRAS-mediated cues. Rocha and colleagues developed complementary principles for LIBRAS-based educational software grounded in semiotic theory [9,10], showing that when images, written language, and signs are combined in educational software, their value depends on whether learners can connect those resources during activity, not on whether each is individually present. Applied to LIBRAS Academy, this principle means that a musical note, a color, a handshape, a written label, and a feedback message form a network: accessibility depends on the coherence of the whole arrangement, not only on the presence of LIBRAS content. Neither Benites nor Rocha, however, evaluated how deaf adolescents interact with such an interface in first-contact conditions, a gap this study addresses.

The broader HCI literature on digital technologies for deaf and hard-of-hearing (DHH) learners explains why that gap matters. A recent systematic review of over a hundred papers on DHH technologies identified language learning, communication support, and family sign language practice as the dominant research areas; music education and engagement-based interfaces for deaf learners remain underrepresented [13]. Existing sign-language applications indicate that their value cannot be assumed from design intentions: Chuan and Guardino [4] demonstrated that even well-designed ASL applications for DHH children and families depend on whether users can coordinate signs, feedback, visual prompts, and task goals in context, an empirical question that design alone cannot answer. More fundamentally, Korte et al. [6] showed that requirements and usability judgments for deaf children cannot be inferred from adult or hearing-user expectations: language development, communication mediation, and ways of expressing difficulty all shape how children interpret and respond to a system. The same concern applies to post-use evaluation: instruments and protocols designed for hearing adults may produce unreliable or uninterpretable responses with DHH populations when applied without adaptation.

Measuring engagement provides a structured way to complement behavioral observation, but raises its own challenges in this context. The User Engagement Scale (UES), developed by O'Brien and Toms [8], operationalizes engagement as a multidimensional construct covering Focused Attention, Perceived Usability, Aesthetic Appeal, and Reward. These dimensions capture aspects of perceived experience that behavioral observation alone cannot: whether a user felt in control, found the interface visually appealing, or considered the activity worthwhile. The UES-Br adaptation by Miranda, Li, and Darin [7] provides a validated Portuguese-language version for Brazilian HCI contexts, and was selected here because using an English-language instrument with Brazilian adolescents would introduce an avoidable validity threat. However, UES-Br was validated with adult Brazilian participants; it has not been applied with deaf populations of any age, and several of its items, such as “I let myself be carried away” and “gratifying,” involve idiomatic or abstract vocabulary that may be difficult regardless of the language of administration. Doherty and Doherty argue that engagement research should use a diverse methodological toolkit rather than treating any instrument as a direct window onto the phenomenon itself [5]. That argument shapes the analysis here: UES-Br responses are treated as evidence of perceived experience, observational records as evidence of behavior, and their convergence or divergence is informative because the two capture different phenomena.

The challenge of eliciting reliable self-report from deaf adolescents runs through all of the above. Young learners' judgments of their own experience are shaped by affect, social context, available vocabulary, and the conditions under which a question is asked, not only by the interaction itself [11,12]. For DHH learners, these conditions are more demanding: spoken think-aloud, self-administered questionnaires in written form, and post-task interviews all assume communication conditions that may not hold for participants responding through sign language, limited written Portuguese, or adult mediation [1]. There is limited

guidance for conducting accessible UX evaluation with deaf adolescents in sign-language-mediated settings. The present study addresses that gap by combining behavioral observation, a mediated UES-Br administration, and open-ended comments, and by reporting what participants did and said alongside the conditions under which each form of evidence was produced.

3 Methodology

Study design. This is a qualitative exploratory study that characterizes how six deaf adolescents interacted with LIBRAS Academy during individual first-contact sessions, and how their perceived engagement, reported through UES-Br, can be triangulated with behavioral observation. The design prioritizes description of first-contact interaction over measurement of outcomes or generalization.

Setting. All sessions were conducted on April 6, 2026, at the Children’s Hearing Center (CeAC), within DERDIC, at PUC-SP, in São Paulo, Brazil. DERDIC is a specialized educational and clinical center for deaf education, hearing and language rehabilitation, LIBRAS support, and family services. This setting provides access to deaf adolescents in a context where sign language use, family mediation, and inclusive education practices are part of the institutional routine.

Participants. Six deaf adolescents aged 13 to 15 were recruited through the institution. Inclusion criteria were: being a student or service user of DERDIC or CeAC, having at least some familiarity with LIBRAS, and having a responsible adult present during the session. Participants had not previously used LIBRAS Academy. They are identified by codes P1 to P6 to preserve anonymity. Table 1 summarizes their profiles; differences in hearing history, LIBRAS background, prior music experience, and written Portuguese fluency are relevant because they shaped how each participant interpreted instructions, managed task demands, and responded to the post-use questionnaire.

Session protocol. Each session was conducted individually and followed a structured script. The research team comprised three members: one researcher dedicated to interacting with the participant and guiding the task sequence, one researcher responsible for observational annotation and audiovisual documentation, and a LIBRAS interpreter who translated task instructions and questionnaire items when needed. A responsible adult was present in all sessions. Sessions began with a profile interview answered by the responsible adult and, when possible, by the participant, covering hearing history, LIBRAS background, prior music and technology experience. The task block comprised seven activities presented in fixed order: free exploration of the home screen; “Alphabet in LIBRAS”; “What Are the Musical Notes?”; “Conductor of Notes”; “Musical Rainbow”; “Note Collector”; and a free-choice final task. Participants could request help at any moment and could indicate readiness to advance to the next activity. Total session duration ranged from 7 to 18 minutes across participants.

Data collection. The empirical corpus comprises: manual observational annotations recorded in real time by the documentation researcher; task pro-

Table 1. Participant profiles. HS = high school year; gr. = grade.

Code	Age / year	Hearing profile	LIBRAS	Music / Portuguese
P1	14; 1st HS	Moderate hearing loss	Since 2015 (11 yrs)	No music study; standard written Portuguese fluency.
P2	14; 1st HS	Complete deafness; no variation	Learned in deaf family (parents deaf)	No music study; standard written Portuguese fluency.
P3	14; 9th gr.	Asymmetric (right 15%, left none)	4–5 years; deaf cousins	Some prior music exposure; standard fluency.
P4	13; 9th gr.	Complete deafness; prior hearing variation	Since childhood (5 yrs)	About 1 year of music study; standard fluency.
P5	15; 9th gr.	Partial hearing (left ear)	3 years; deaf uncle	No formal music study; limited written Portuguese.
P6	15; 1st HS	Partial hearing (right ear)	11 years; only deaf in family	Likes music; stronger written Portuguese fluency.

gression notes indicating which phases were reached in each activity; mediation records noting when and how support was provided; and post-use responses to an adapted administration of UES-Br exported from Google Forms. Unlike the validated self-administered format, items were presented one by one in Portuguese and mediated in LIBRAS by the interpreter as needed; participants responded at their own pace, and responses were entered into Google Forms by the participant or, when needed, by the interpreter. This deviation from the validated protocol was a deliberate choice to make the instrument accessible; its consequences for response validity are examined in the results. Screen recordings were collected for documentation and event verification; full audiovisual coding was beyond the scope of this study and is planned for future work.

UES-Br scoring. The instrument contains 30 Likert items (scale 1–5) grouped into four dimensions: Focused Attention (FA, 7 items), Perceived Usability (PU, 8 items), Aesthetic Appeal (AE, 5 items), and Reward (RW, 10 items), plus one open-ended prompt. FA and AE contain only positively worded items and were computed as item means. PU contains seven negatively worded items (PU1–PU6 and PU8) and one positively worded item (PU7); negatively worded items are inverted using $\text{score}_{\text{corr}} = 6 - \text{score}_{\text{raw}}$ before averaging. RW3 is negatively worded and was likewise inverted before computing the RW mean. Both dimensions are reported as corrected means in Table 3; raw values are cited in the text where needed for comparison. All individual item responses were verified by manual inspection against the original PDF exports of the Google Forms records prior to computing means. Given the small sample and the mediated adminis-

tration, UES-Br results are used descriptively and as complementary evidence of perceived experience, not for inferential claims.

Qualitative analysis. Observational annotations were coded using a hybrid thematic procedure. An initial set of codes was derived deductively from the three research questions; additional codes were developed inductively through close reading of the annotation corpus. The coding was performed by one researcher and reviewed in a collaborative session with the full research team, where codes were consolidated and discrepancies resolved by consensus. The final codebook comprised five categories: interface barriers (layout, pace, recognition, scale); persistence and voluntary continuation; curiosity and exploratory behavior; instrument accessibility problems; and mediation and social context. Coded events were organized by participant and activity to support cross-case comparison. The analysis does not reach theoretical saturation; it identifies recurring patterns and contrastive cases across six first-contact sessions. In the triangulation, UES-Br responses are treated as evidence of perceived experience, and observational codes as evidence of behavior; their convergence or divergence is informative.

Ethical considerations. The study followed institutional procedures for research activities involving minors. Informed consent was obtained from responsible adults; participants provided verbal assent. Participation was voluntary; participants could withdraw at any time without penalty. Anonymity is preserved through participant codes P1–P6; no names, school identifiers, or other direct identifiers appear in the analysis or reporting.

4 Results

4.1 Interaction Trajectories Across Sessions

The six sessions followed the same activity sequence, but they did not produce a single engagement profile. Each participant interpreted and used the interface in a different way: through reflective inquiry, efficient completion, self-monitoring, selective persistence, image-based navigation, or autonomous interpretation. Duration ranged from 7 minutes and 25 seconds (P4) to 18 minutes and 9 seconds (P5). That range does not reflect engagement level: P4, the fastest, persisted in one activity until the progress bar was full; P5, the longest, showed reading difficulty and possible social desirability in questionnaire responses. Table 2 maps the dominant interaction condition observed for each participant across the seven activities (● autonomous; △ local mediation; × structural barrier).

Reflective inquiry. P1 asked whether the tool was Brazilian or American, spelled out his own name in the alphabet game, questioned the double meaning of “Sol” (a musical note and the word for the sun) and entered the computer’s system settings to comment on its speed relative to his own machine. Such questions show that engagement with a sign-language educational tool extends to its cultural origin, linguistic assumptions, and technical infrastructure. His session was the second longest (14:36), not because of difficulty, but because inquiry was part of how he participated.

Table 2. Dominant interaction pattern by participant and activity. • = predominantly autonomous; Δ = local mediation at task onset or at a specific moment; $\times$ = structural barrier that disrupted the activity regardless of explanation.

Part.	Exp.	Alph.	Notes	Cond.	Rainb.	Coll.	Free	Distinguishing characteristic
P1	•	•	Δ	Δ	Δ	•	•	Inquiry into the cultural, linguistic, and technical conditions of the interface.
P2	•	Δ	•	Δ	$\times$	Δ	Δ	Efficient task completion; pace and layout barriers; limited voluntary continuation.
P3	•	Δ	•	•	•	•	•	Most autonomous session; self-correction during free-choice play.
P4	•	Δ	•	•	Δ	•	•	Fastest session; pace-sensitive; persisted in a progress-based activity.
P5	Δ	Δ	Δ	•	$\times$	•	Δ	Image-based navigation; reading difficulty; scroll barrier; possible social desirability.
P6	•	$\times$	Δ	•	•	•	•	Autonomous reading; system recognition failure for letter O.

Efficient completion without voluntary extension. P2 grasped activity goals quickly and executed them with little disruption, recalling Do, Re, Mi before identifying Fa and Sol and progressing through the alphabet with localized difficulty on specific letters. Yet efficiency did not carry over into continuation: at the free-choice stage he did not want to explore further and proceeded only after encouragement. An interface can be navigable without being the kind of experience a participant would choose to extend.

Self-monitoring and repair. P3’s session was the most consistently autonomous. She moved through long activity sequences with support only on specific alphabet letters and encountered no structural barriers across the music tasks. This was most visible in the free-choice stage: during the memory game she made an error, recognized it, and corrected it without assistance, then continued in “Build the Word” with sustained performance. Noticing a mistake, evaluating it, and correcting it requires using the interface’s feedback as information, not merely responding to prompts.

Selective persistence. P4 completed the sequence faster than any other participant. Her impatience with slower tasks was observable: she did not wait for “Conductor of Notes” to finish and encountered difficulty in “Musical Rainbow” when her pace exceeded the interface’s feedback rhythm. Yet she remained with “Note Collector” until the progress bar was fully completed. She disengaged from rhythm-dependent tasks but committed to a progress-based one, which makes

the task, rather than the whole session, the more useful unit for engagement analysis.

Image-based navigation. P5 oriented strongly toward images and colors rather than written labels, needed initial support to activate the camera, and then progressed through the music tasks using color-to-note associations as his anchor. His session was the longest (18:09), though the observer noted a possible desire to please the researchers, a factor that bears on how his questionnaire responses should be read.

Autonomous interpretation and system limitation. P6’s stronger written Portuguese fluency allowed him to read interface instructions without consistent mediation. His difficulties were localized and produced by the system rather than by his understanding: a recognition failure for the letter O (a correct handshape the gesture system did not register) and a brief hesitation around Sol. When linguistic access to written content is strong, the system’s own recognition limitations become the obstacle that matters most.

4.2 Interface Access Barriers

Most mediation was local: once an activity rule or expected action was clarified, participants could proceed. The more consequential barriers were those that explanation alone could not resolve.

Visual layout and scale. P2 had initial difficulty in “Musical Rainbow” because notes were distributed across different vertical positions rather than aligned on a shared line, disrupting the visual mapping between sequence and action. P1 required a zoom adjustment because the default scale was too small for comfortable use. These barriers reflect layout decisions that assume a visual context the interface does not consistently provide.

Pace and temporal control. P2 perceived “Note Collector” as too slow and tried clicking to accelerate it. P4 was asked by the mediator to slow down in “Conductor of Notes,” and her haste in “Musical Rainbow” interacted with the task’s requirement for deliberate sequencing. The ability to match or adjust an activity’s rhythm is an accessibility dimension independent of LIBRAS content presence.

Recognition and input reliability. P5’s scroll interaction made the “see again” button in “Musical Rainbow” unreachable, interrupting an activity that depends on replay. P6’s correct handshape for the letter O was not recognized by the gesture system. In both cases the barrier originated in the system, not in the participant’s competence or effort.

Accessible design for sign-language-mediated music learning therefore requires visual coherence, temporal flexibility, and recognition reliability, not only the presence of LIBRAS content.

Table 3. Descriptive UES-Br means by participant (corrected scores). PU and RW negatively worded items were inverted ($6 - \text{score}$) before averaging. Raw means are reported in the text where necessary for comparison.

Code	FA	PU	AE	RW
P1	3.86	3.25	3.00	3.50
P2	3.00	3.12	3.80	3.80
P3	3.43	2.62	3.80	4.00
P4	2.29	4.12	1.80	2.70
P5	3.43	4.88	5.00	4.90
P6	3.29	2.88	3.60	3.30

4.3 UES-Br as Complementary Evidence

The UES-Br data in Table 3 did not simply confirm or contradict the observations. It clarified some cases, complicated others, and exposed constraints in the evaluation procedure itself.

Where the data clarified, it added a self-report dimension that observation alone cannot provide. P3’s fluent, self-correcting interaction is accompanied by the strongest reward scores in the dataset ($RW = 4.00$) and consistent aesthetic appeal; the questionnaire is consistent with the interpretation that what appeared behaviorally as engagement was also experienced as worthwhile and visually appealing. P6’s autonomous, mostly unmediated interaction corresponds to moderate, coherent scores across all dimensions. In these two cases, the two sources of evidence tell a consistent story.

Where the data complicated, it revealed that behavioral evidence and self-report can point in different directions, and both directions are real. P4 presents the sharpest discrepancy: the lowest focused attention ($FA = 2.29$) and aesthetic appeal ($AE = 1.80$) in the dataset, while the observational record indicates that she stayed with “Note Collector” until the progress bar was full. The scores describe low absorption; the behavior describes goal-directed commitment to a specific task. P1 completed the activities with local mediation but required extensive explanation for questionnaire expressions such as “I let myself be carried away,” “aesthetically pleasing,” and “gratifying,” expressions the interface itself never demanded from him. His scores document a difficult questionnaire administration, not a difficult interaction. P2 combined efficient completion with limited voluntary continuation, yet his reward scores are among the higher in the group ($RW = 3.80$), and his open response in Portuguese (“Eu achei de toda experiência, essa apresentou o jogo ser importante tentar em libras e criado em libras!”; translated by the authors as: “I found that the experience showed the importance of trying a game in LIBRAS, created in LIBRAS”) is the most explicit endorsement of the platform. Each of these signals captures something the others do not.

Scoring direction also matters for interpretation. Most PU items describe frustration, confusion, or difficulty; high raw agreement therefore signals worse

usability, not better. P5’s raw PU mean of 1.62 (very low agreement with negative statements) becomes a corrected score of 4.88, the highest in the group. P3 and P6 move in the opposite direction: their raw means of 3.62 and 3.38 reflect moderate agreement with negative items, yielding corrected scores of 2.62 and 2.88. Reporting uncorrected values would have inverted the interpretation of all three cases.

Where the data exposed constraints, it revealed specific conditions under which UES-Br responses in this context should not be taken at face value. Three threats emerged. The first is *semantic inaccessibility*: items such as “I let myself be carried away,” “aesthetically pleasing,” and “gratifying” required negotiation with the interpreter before participants could respond, so the resulting score reflects that negotiation alongside the original experience. The second is *social desirability*: P5 produced the highest corrected PU, AE, and RW in the dataset in a session the observer described as showing reading difficulty, image-based orientation, and apparent disinterest in some tasks. Treating those scores as direct evidence of high engagement would misrepresent the observed session. The third is *mediation effect*: when an interpreter or responsible adult presents items and records answers, the relational dynamic changes the social meaning of agreement. These three threats do not invalidate UES-Br as an instrument; they define the conditions under which its scores in this study require triangulation rather than direct interpretation.

5 Discussion

The results produce three specific answers. First-contact engagement with LI-BRAS Academy was not a single interaction profile (RQ1): participants built access to the interface through six distinct paths, including inquiry, efficient completion, self-correction, selective persistence, image-based navigation, and autonomous reading. Triangulating behavioral observation with UES-Br scores was productive precisely because the two sources diverged (RQ2): P4’s low attention scores coexisted with goal-directed persistence; P1’s successful task interaction coexisted with questionnaire difficulty; P2’s efficient completion coexisted with an enthusiastic endorsement of the platform. UES-Br responses were also subject to three interpretability threats that required triangulation rather than direct interpretation (RQ3).

The results also refine claims from the related literature. Korte et al. [6] showed that usability requirements for deaf children cannot be inferred from adult or hearing-user expectations; our results extend that argument to post-use evaluation: engagement instruments, not only interface requirements, need adaptation for this population. Chuan and Guardino [4] demonstrated that the value of sign-language applications depends on whether users can coordinate signs, feedback, visual prompts, and task goals in context. The barriers observed here specify what that coordination failure looks like: P2’s layout barrier disrupted the visual chain between note position and expected action; P4’s pace mismatch disrupted the temporal chain between the interface’s feedback rhythm and the

learner’s anticipation; P5’s scroll barrier removed the ability to use the interface’s replay function for self-regulation; P6’s recognition failure broke the correspondence between a correct gesture and the system’s response. None of these is a localized defect in the ordinary usability sense. Each is a breakdown in a chain of interdependent resources (LIBRAS sign, color, musical note name, gesture, feedback, written label) where access depends on whether the chain holds. This shifts the design question from the presence of LIBRAS content to the coherence of the relations among signs, visual prompts, written labels, gestures, and feedback.

From this perspective, the six engagement trajectories are not six individual differences in motivation or ability. They reflect six ways of building that legibility under different language and mediation conditions. P6 could read written labels directly and so encountered the system’s recognition limits as his most immediate barrier; P5 could not rely on written labels and so built legibility through color-based associations, a path that worked until a scroll problem made replay inaccessible. P3 could monitor her own errors and use feedback to correct them; P2 could complete tasks without choosing to continue voluntarily. Understanding these differences matters for inclusive educational technology beyond LIBRAS Academy: it indicates that accessible design must account for the multiple paths through which different users construct access to the same interface, not only for the presence of accessibility features.

The UES-Br results have a methodological implication for the field. The three interpretability threats identified are not specific to this instrument or to this study: they describe conditions that may arise in mediated self-report evaluation with deaf adolescents. Semantic inaccessibility can affect instruments that use idiomatic or abstract vocabulary without items adapted and validated for LIBRAS-mediated administration. Social desirability can affect evaluations conducted by researchers in institutional settings where participants have limited power to express negative assessments. Mediation effect can affect protocols in which a third party reads and records the participant’s response. Together, they indicate that accessible UX evaluation with deaf adolescents cannot be reduced to translating an instrument into Portuguese or providing a LIBRAS interpreter. It requires rethinking how items are formulated, how the evaluation is socially positioned, and how responses are recorded. The goal is an evaluation arrangement designed around the participant’s language access, not an instrument administered after use.

6 Limitations and Future Work

Three constraints define the scope of the study’s claims. The sample comprises six participants tested in a single first-contact session at one institution, which means the results cannot be generalized statistically and should be read as exploratory evidence about what engagement with LIBRAS Academy looked like under these specific conditions. This sample size is consistent with exploratory qualitative HCI work with specialized populations that prioritize depth of de-

scription over breadth [6,4], but it limits the study to analytic rather than statistical generalization. The corpus relies on manual observational annotations and mediated UES-Br responses rather than a fully coded audiovisual record, which limits the precision with which behavioral events can be reconstructed. Researcher presence, institutional familiarity, and mediated questionnaire administration shaped what was measured alongside what was experienced; the study cannot cleanly separate engagement from the social conditions of its observation.

An additional constraint is the novelty effect inherent in first-contact design: all participants encountered LIBRAS Academy for the first time, so the patterns described reflect discovery and initial orientation rather than typical or sustained use.

These constraints point to three directions that follow directly from what was found. The six engagement trajectories identified here raise the question of which are systematic and which are idiosyncratic to these six individuals; additional sessions with a broader and more varied participant set are needed to answer it. The observation that first-contact performance does not index sustained engagement or learning motivates longitudinal work: repeated sessions could examine whether participants become more autonomous over time, whether barrier categories shift as familiarity grows, and whether LIBRAS-mediated music activities contribute to outcomes beyond music itself. Finally, the three interpretability threats identified for UES-Br (semantic inaccessibility, social desirability, mediation effect) and the three interface barrier categories (visual layout, temporal control, recognition reliability) define a practical agenda for the next study: address the instrument before re-administering it, and address the interface before re-testing it.

7 Conclusion

First-contact engagement with a sign-language-mediated music learning interface is not a single experience. Across six deaf adolescents, it took six behaviorally distinct forms, none reducible to the others, and none fully captured by a single score or by task completion alone. Beyond LIBRAS Academy, this means that accessible educational technology for deaf adolescents in sign-language-mediated contexts should be evaluated through what users do with the available resources. Such evaluation requires methods designed around the population’s language access, not only instruments adapted from hearing-user research.

Methodologically, UES-Br produced interpretable evidence in this study, but only when scores were corrected for negatively worded items, read alongside observational records, and checked against the conditions under which each response was produced. Three interpretability threats must be addressed before engagement can be measured more reliably with deaf adolescents: semantic inaccessibility of questionnaire vocabulary, social desirability under researcher presence, and the relational effect of mediated administration.

Together, the six trajectories, the barrier categories, and the UES-Br interpretability threats provide a starting point for future work. Broader participation, repeated sessions, interface refinement, and instrument adaptation are the next steps.

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